

AI Product Intelligence System Report

1. Overview

This project extends a multimodal AI-based Product Intelligence System using CLIP embeddings. The system performs product recommendation, catalog deduplication, and reverse product search using image and text embeddings to optimize discovery, cross-selling potential, and catalog tracking operations across large scale e-commerce applications.

2. Technical Implementation & Task Breakdowns

Task 1: Smart Product Recommendation Engine

A recommendation engine was developed to suggest complementary products based on product categories, allowing e-commerce platforms to improve cross-selling opportunities efficiently.

- **Input Product:** Sports Shoes
- **Recommended Products:** Sports Socks, Fitness Watch, Water Bottle

Task 2: Unique Product Catalog Creation

CLIP image embeddings and density-based DBSCAN clustering were implemented to identify duplicate and near-duplicate products submitted by conflicting sellers, grouping vectors automatically and creating a clean master catalog using centroid proximity filtering.

- **Original Catalog Size:** 500
- **Unique Catalog Size:** 270

Task 3: Reverse Product Search

A text-based search system was implemented using CLIP text embeddings and cosine similarity scores. This allows users to search products using natural language instead of uploading reference images.

- **Example Query:** blue casual shirt
- **Top Matching Products:** 57281.jpg, 25540.jpg, 13333.jpg, 4772.jpg, 12267.jpg

3. System Screenshots & Execution Output

The system execution completed successfully with the following terminal outputs captured across all verification modules:

```
===== TASK 1 =====
Input Product: Sports Shoes

Recommended Products:
- Sports Socks
- Fitness Watch
- Water Bottle

===== TASK 2 =====
Original Catalog Size: 500
Unique Catalog Size: 270

===== TASK 3 =====
Query: blue casual shirt
1. 57281.jpg | Score=0.2998
2. 25540.jpg | Score=0.2963
3. 13333.jpg | Score=0.2952
4. 4772.jpg | Score=0.2932
5. 12267.jpg | Score=0.2924

===== SUMMARY =====
Task 1 Completed ✓
Task 2 Completed ✓
Task 3 Completed ✓

Notebook Execution Complete
```

4. Technologies Used

Python

OpenCLIP

PyTorch

Pandas

NumPy

DBSCAN

Cosine Similarity

Matplotlib

5. Zip File Submission Directory

As requested by the submission guidelines, the final archive contains the following structural mapping:

```
Submission.zip
|
├── AI_Product_Intelligence_System.ipynb
└── Report.pdf
```

6. Conclusion

The AI Product Intelligence System successfully implemented and benchmarked all three advanced multimodal vector pipeline features. The project directly demonstrates the immense practical utility of joint image-text model spaces for automating manual operational workflows within modern, large scale e-commerce applications.